

Abstract

The CKM and PMNS flavor mixing matrices of the Standard Model lack a first-principles geometric origin. We show that if fermion flavors are associated with distinct boundary directions of a compact hyperbolic 3-manifold, the resulting mixing matrices are uniquely determined — up to sector-local rephasings — by the boundary configuration alone, and form a *discrete* set in $U(n)/(U(1)^n \times U(1)^n)$. Our main result (Theorem 10) establishes that no continuous deformation of physical parameters can produce continuous drift in any rephasing-invariant mixing observable. The construction is purely kinematic, requiring no dynamical input, and provides a model-independent structural constraint on any geometric theory of flavor mixing. In companion papers [10, 11], this framework is applied to two specific arithmetic hyperbolic 3-manifolds, reproducing the observed CKM and PMNS mixing matrices with Frobenius fitness 0.016482 and 0.005087 respectively.

Discrete Mixing Operators from Boundary Sector Geometry

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1 Introduction

The CKM quark mixing matrix [1, 2] and the PMNS lepton mixing matrix [4, 3] encode the mismatch between flavor and mass eigenstates. Despite their precise experimental determination [5], their values have no derivation from the gauge structure of the Standard Model. Discrete flavor symmetries [6], extra dimensions, and string-inspired constructions [7] have all been proposed, but a kinematic, model-independent explanation has been lacking.

In this paper we take a minimal perspective: fermions are assigned to distinct boundary directions $\xi(f) \in S^2 = \partial\mathbb{H}^3$ of a compact hyperbolic 3-manifold $M = \mathbb{H}^3/\Gamma$. The mixing matrix between any two sectors arises as the overlap operator between the corresponding boundary-localized Hilbert subspaces. We prove that under a discrete assignment rule, the set of admissible mixing matrices is finite modulo rephasings — so continuous drift in any rephasing-invariant observable is geometrically forbidden.

This result is the kinematic foundation for the Hyperbolic Flavor Geometry (HFG) programme [12]. In the companion CKM paper [10], the K-factor of the Iwasawa decomposition $\mathrm{PSL}(2, \mathbb{C}) = KAN$ applied to the holonomy of $M_{\mathrm{CKM}} = m006(-5, 2)$ gives Frobenius fitness 0.016482 against PDG 2024 values. In the companion PMNS paper [11], the N-factor (Borel) applied to $M_{\mathrm{PMNS}} = m003(-2, 3)$ gives fitness 0.005087 — the global minimum of the construction. The present paper provides the mathematical guarantee that these results are not accidents of continuous parameter tuning: the admissible mixing matrices form a discrete set.

2 Geometric Setup

Let Γ be a torsion-free discrete group acting properly discontinuously and cocompactly on $\mathbb{H}^3$ by orientation-preserving isometries. The quotient $M = \mathbb{H}^3/\Gamma$ is a compact oriented hyperbolic 3-manifold with boundary at infinity $S^2 = \partial\mathbb{H}^3$.

Definition 1 (Boundary direction). A *boundary direction* is a point $\xi \in S^2$. For $\gamma \in \Gamma$, the associated boundary direction is the ideal endpoint of the geodesic ray from a fixed basepoint $o \in \mathbb{H}^3$ through $\gamma \cdot o$.

Assumption 2 (Discrete sector assignment). Each fermion flavor $f \in \mathcal{F}$ is assigned a unique boundary direction $\xi(f) \in S^2$ by a deterministic, discrete selection rule. The image $\Xi = \{\xi(f) : f \in \mathcal{F}\}$ is finite, and the assignment is locally rigid: distinct flavors map to distinct directions, and small perturbations of the basepoint do not change the assignment class.

Remark 3 (Geometric realization). A natural realization: for each flavor f , let $\gamma_f \in \Gamma$ be the unique element minimizing $d(o, \gamma \cdot o)$ in the assignment class of f . Proper discontinuity of the

action implies the orbit $\Gamma \cdot o$ is discrete, so the minimizer exists; torsion-freeness forces uniqueness. The boundary direction $\xi(f)$ is then the ideal endpoint of the geodesic from o through $\gamma_f \cdot o$. In the HFG application, γ_f is a short word in $\pi_1(M)$ and $\xi(f) = \hat{n}(\gamma_f)$ is the geodesic axis direction extracted via the Pauli decomposition of $\log \rho(\gamma_f)$.

3 Boundary Hilbert Space and Sector Subspaces

Let $\mathcal{H}$ be a separable complex Hilbert space carrying boundary data (concretely, $L^2(S^2)$).

Definition 4 (Boundary-localized state). For each $\xi \in S^2$, fix a normalized vector $|\xi\rangle \in \mathcal{H}$ such that $\xi \mapsto |\xi\rangle$ is continuous, $\langle \xi | \xi \rangle = 1$, and $|\langle \xi | \zeta \rangle| < 1$ for $\xi \neq \zeta$. An explicit family is the Poisson kernel: $|\xi\rangle \propto P(\cdot, \xi)$ normalized in $L^2(S^2)$, which satisfies all three conditions [9].

Definition 5 (Sector and sector subspace). A *sector* S is a finite ordered subset of $\mathcal{F}$. The *sector subspace* is $\mathcal{H}_S = \text{span}\{|\xi(f)\rangle : f \in S\} \subset \mathcal{H}$, with $\dim \mathcal{H}_S = |S|$ (since distinct boundary directions give linearly independent states). The *preferred ONB* $\{|S, 1\rangle, \dots, |S, n\rangle\}$ of $\mathcal{H}_S$ is the Gram-Schmidt orthonormalization of $(|\xi(f_1)\rangle, \dots, |\xi(f_n)\rangle)$ with all pivot coefficients real and positive, removing the $U(1)^n$ phase ambiguity.

4 Mixing Matrices as Overlap Operators

Definition 6 (Mixing matrix). For sectors S, T of equal cardinality n , the *mixing matrix* U_{ST} is the $n \times n$ matrix with entries $(U_{ST})_{ji} = \langle T, j | S, i \rangle$.

Proposition 7 (Unitarity and geometric dependence). When $\mathcal{H}_S = \mathcal{H}_T$, $U_{ST} \in U(n)$. In general it is a partial isometry. U_{ST} is determined entirely by the finite boundary configuration $\Xi_{ST} = \{\xi(f) : f \in S \cup T\}$ and the inner product on $\mathcal{H}$: all Gram-Schmidt coefficients are algebraic expressions in the inner products $\langle \xi(f) | \xi(g) \rangle$, $f, g \in S \cup T$.

5 Rephasing Invariance

Physical observables are invariant under sector-local rephasings: $U_{ST} \rightarrow D_T U_{ST} D_S^\dagger$, where $D_S, D_T \in U(1)^n$. The physical content is the equivalence class of U_{ST} in $U(n)/(U(1)^n \times U(1)^n)$.

Proposition 8 (Rephasing-invariant quantities). The following are invariant under sector-local rephasings: (i) the moduli $|(U_{ST})_{ji}|^2$ (mixing probabilities); (ii) the quartic products $J_{ji,j'i'} = (U_{ST})_{ji}(U_{ST})_{j'i'}(U_{ST})_{ji'}^*(U_{ST})_{j'i}^*$ (generalizing the Jarlskog invariant [8]); (iii) $\det(U_{ST})$ up to an overall phase. For $n = 3$ and the standard CKM/PMNS conventions, $\text{Im}(J_{ji,j'i'})$ reduces to the standard Jarlskog invariant J_{CP} .

6 Discreteness and Stability

Theorem 9 (Stability under deformations). Let $\delta_{\min} = \min_{f \neq f'} d_{S^2}(\xi(f), \xi(f'))$ be the minimal pairwise separation of boundary directions, and set $\varepsilon_0 = \delta_{\min}/2$. For any deformation of magnitude $\varepsilon \leq \varepsilon_0$ (each $\xi(f)$ moved by at most ε in the round metric on S^2), the perturbed configuration determines the same preferred bases and the same mixing matrix U_{ST} .

Proof. For $\varepsilon \leq \varepsilon_0$, each perturbed point $\xi'(f)$ remains closer to $\xi(f)$ than to any other $\xi(g)$, $g \neq f$, so the points remain distinct and carry the same labels. Linear independence is therefore preserved. The Gram-Schmidt coefficients depend continuously on the inner products $\langle \xi'(f) | \xi'(g) \rangle$, which vary continuously with ε ; but the discrete assignment rule (Assumption 2) is unchanged since each $\xi'(f)$ belongs to the same assignment class as $\xi(f)$. By Proposition 7, U_{ST} depends only on the configuration class, which is unchanged. $\square$

Theorem 10 (Discreteness of mixing classes). *Under Assumption 2, the set of accessible mixing matrices*

$$\mathcal{M}_{ST} = \{U_{ST} : \Xi_{ST} \text{ admissible}\}$$

is discrete (in fact finite) in $U(n)/(U(1)^n \times U(1)^n)$.

Proof. The set of admissible configurations Ξ_{ST} is finite by Assumption 2. By Proposition 7, the map $\Xi_{ST} \mapsto U_{ST}$ is continuous, hence maps a finite set to a finite set. Quotienting by the rephasing action (a continuous group action on a finite set) preserves finiteness. Thus $\mathcal{M}_{ST}$ is a finite, hence discrete, subset of $U(n)/(U(1)^n \times U(1)^n)$. $\square$

Corollary 11 (No continuous drift). *For any continuous deformation of boundary directions of magnitude $\varepsilon \leq \varepsilon_0$, no continuous drift occurs in any rephasing-invariant mixing observable. For larger deformations that change the configuration class, the mixing matrix can change, but only by a discrete jump to another element of $\mathcal{M}_{ST}$.*

Proof. Rephasing-invariant observables are continuous functions on $U(n)/(U(1)^n \times U(1)^n)$. Their restriction to the finite set $\mathcal{M}_{ST}$ takes only finitely many values; continuous drift is therefore impossible. $\square$

7 Connection to Hyperbolic Flavor Geometry

The abstract framework above is concretely realized in the HFG programme [12] as follows. The group $\Gamma = \pi_1(M)$ acts on $\mathbb{H}^3$ via the holonomy representation $\rho : \pi_1(M) \rightarrow \text{PSL}(2, \mathbb{C})$. For a loxodromic element γ , the boundary direction $\xi(\gamma) = \hat{n}(\gamma) \in S^2$ is the geodesic axis, extracted via the Pauli decomposition of $\log \rho(\gamma)$. The inner product $\langle \xi(f) | \xi(g) \rangle$ becomes the Gaussian kernel $\exp(-\theta_{fg}^2/(2\sigma^2))$, where $\theta_{fg} = \arccos |\hat{n}(f) \cdot \hat{n}(g)|$.

For $M_{\text{CKM}} = m006(-5, 2)$ (OrientableClosedCensus[43], $H_1 = \mathbb{Z}/5$, $\text{vol} = 2.029$), the three-word sector $\{\text{aaB}, \text{AbA}, \text{AAb}\}$ gives a mixing matrix with Frobenius fitness 0.016482 against PDG 2024 CKM values, and Cabibbo angle error 0.19%. The homological class collision $[\text{AbA}] = [\text{AAb}] = 2$ in $H_1 = \mathbb{Z}/5$ forces $J_{\text{CP}} = 0$ via Corollary 11 — consistent with the observed suppression $|J_{\text{CKM}}| \approx 3 \times 10^{-5}$.

For $M_{\text{PMNS}} = m003(-2, 3)$ (OrientableClosedCensus[1], $H_1 = \mathbb{Z}/5$, $\text{vol} = 0.981$), the Borel (lower-triangular) construction on the sector $\{\text{aa}, \text{aaB}, \text{baa}\}$ gives fitness 0.005087 — the global minimum of the construction, with $p < 10^{-4}$ against 50,000 Haar-random unitary matrices. The distinct homology classes $[\text{aa}] = 1$, $[\text{aaB}] = 4$, $[\text{baa}] = 3$ (all distinct mod 5) ensure $J_{\text{CP}} \neq 0$, consistent with the large leptonic CP violation observed.

8 Conclusions

We have shown that the discreteness of boundary directions in a compact hyperbolic 3-manifold implies the discreteness of all admissible mixing matrices modulo rephasings (Theorem 10), and that bounded deformations cannot produce continuous drift in physical observables (Theorem 9, Corollary 11). This provides a model-independent kinematic explanation for why the CKM and PMNS mixing matrices cannot vary continuously: they are selected from a discrete geometric set determined by the topology of the underlying manifold.

Data Availability

No data were generated. Companion computational scripts are at <https://github.com/drmlgentry/hyperbolic-flavor-scan>.

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